Do not write outside the box

**0 4** Fertilisers are formulations.

**0 4 . 1** What is a formulation?

**[1 mark]**

_____

_____

**0 4 . 2** A bag of fertiliser contains 14.52 kg of ammonium nitrate ($NH_4NO_3$).

Relative formula mass ($M_r$): $NH_4NO_3$ = 80

Calculate the number of moles of ammonium nitrate in the bag of fertiliser.

Give your answer in standard form to 2 significant figures.

**[4 marks]**

_____

_____

_____

_____

_____

_____

_____

_____

Moles of ammonium nitrate = _____ mol

**Question 4 continues on the next page**

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**0 4 . 3** The fertiliser also contains potassium chloride.

Explain why potassium chloride has a high melting point.

**[4 marks]**

9

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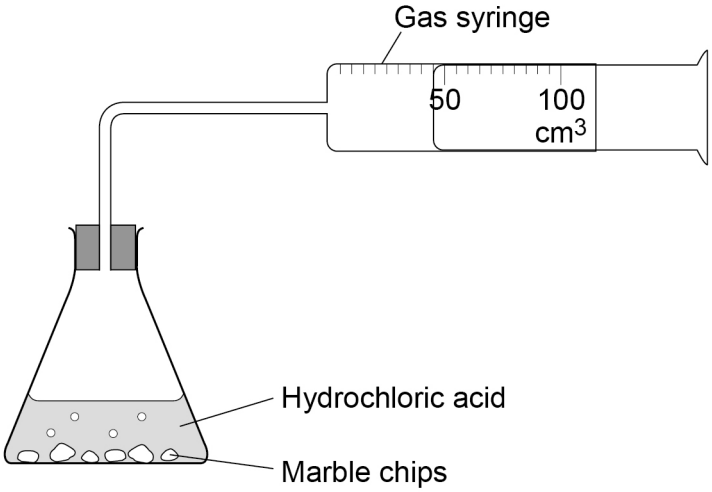
**0 5** A student investigated the effect of the size of marble chips on the rate of the reaction between marble chips and hydrochloric acid.

This is the method used.

1. Add 10 g of marble chips into the flask.
2. Add 50 $cm^3$ of hydrochloric acid, connect the gas syringe and start a timer.
3. Record the volume of gas produced every 10 seconds.

**Figure 4** shows the apparatus.

**Figure 4**




**0 5 . 1** Complete the equation for the reaction.

**[2 marks]**

$CaCO_3$ + ____ HCl → _____

**Question 5 continues on the next page**